WEB3NOVA / AI SKILLS SERIES

AI PROMPT ENGINEERING

A beginner-friendly visual guide to asking AI for better answers, designs, videos, code and ideas.



Understand it • Build it • Improve it

What you will learn

UNDERSTAND

What prompts and prompt engineering mean

WRITE

Build clear prompts with a repeatable structure

COMPARE

Spot why vague prompts often miss the goal

CREATE

Use prompts for text, designs, video and code

First: what is generative AI?

AI MODEL

A system trained on examples that can generate text, images, audio, video or code.

PROMPT

The instruction, question, context or media you provide to guide what the system produces.

OUTPUT

The response or artifact produced. It may be helpful, incomplete, inaccurate or unexpected.

AI is a powerful assistant—not an all-knowing authority.

What is a prompt?

A prompt is the input you give an AI to tell it what you want.

SIMPLE PROMPT

Explain blockchain to a beginner.

WHAT IT TELLS THE AI

Task: explain
Topic: blockchain
Audience: a beginner

Prompts may include text, images, audio or files, depending on the tool.

What is prompt engineering?

Designing, testing and improving instructions so AI outputs better match your goal.

1

1 • DEFINE

What do I need?

2

2 • INSTRUCT

Give clear details

3

3 • REVIEW

Check the result

4

4 • REFINE

Improve one thing

There is no single magic phrase. Clear communication and iteration matter most.

Why the prompt changes the result

VAGUE INPUT

“Make something nice.”

The AI must guess the purpose, audience, style, format and details.

→

CLEAR INPUT

“Create a launch poster for a beginner coding workshop…” plus style and layout details.

→

BETTER FIT

A result closer to your purpose—still needs checking.

Visual comparison: weak vs strong prompt

WEAK PROMPT

INPUT

Make a poster.

- Missing audience and purpose
- No exact text, date or venue
- No style, size or layout
- No constraints

STRONGER PROMPT

INPUT

Design a portrait social poster for a free beginner AI workshop. Use navy and cyan, a clear headline, 3 benefits, date/time placeholders and a visible registration CTA. Keep text readable; leave space for a logo.

Gives the AI a target and boundaries.

The anatomy of a strong prompt

ROLE
What perspective should AI take?

TASK
What should it do?

CONTEXT
Who / what is it for?

DETAILS
What must be included?

FORMAT
How should the answer look?

CONSTRAINTS
What should it avoid or preserve?

Not every prompt needs every part. Add only details that help the task.

Copy-and-use prompt template

PROMPT BUILDER

ROLE: Act as a [relevant role].
TASK: Create / explain [deliverable].
CONTEXT: This is for [audience] to achieve [goal].
DETAILS: Include [must-have points].
FORMAT: Present it as [table / steps / script].
STYLE: Use a [tone / visual style].
CONSTRAINTS: Avoid [things to avoid].
QUALITY CHECK: Check against [criteria].

TIP 1

Replace brackets with real details.

TIP 2

Remove sections that are irrelevant.

TIP 3

Ask questions if key details are missing.

Worked example: from vague to useful

VERSION 1

“Teach me marketing.”

Too broad: no audience, goal, depth or format.

VERSION 2

“Explain digital marketing to a beginner in simple language.”

Better, but still broad.

VERSION 3

“Create a 7-day beginner plan with one 20-minute task per day, free tools and a mini project. Use a table.”

Same topic • better direction • more usable output

Principle 1: be clear and specific

- Name the action: write, explain, summarize, design, compare or debug.
- Name the subject: what is the prompt about?
- Describe what “good” means for this task.
- Use concrete descriptions instead of only mood words.
- Put the most important requirement near the beginning.

EXAMPLE

Instead of “Make it professional” → “Use readable sans-serif headings, strong spacing and a navy/cyan palette.”

Principle 2: give useful context

WITHOUT CONTEXT

“Write an announcement.”

The AI does not know the event, audience, tone, channel or call to action.

WITH CONTEXT

“Write a friendly WhatsApp announcement for beginners about an AI class. Mention 3 learning outcomes and use placeholders for missing logistics.”

Context should be relevant—not a dump of unrelated information.

Principle 3: set boundaries and constraints

LENGTH

“Keep it under 150 words.”

FORMAT

“Return a 5-row table.”

AUDIENCE

“Assume no coding experience.”

INCLUDE

“Add 3 examples and a recap.”

AVOID

“Avoid jargon and invented statistics.”

PRESERVE

“Change only the background; keep the logo.”

Principle 4: request the format you need



The format should make the output easier to use.

Principle 5: show an example when helpful

NO EXAMPLE

Write 3 product descriptions.

WITH EXAMPLE

Write 3 descriptions in this style:

“Lightweight. Reliable. Ready for your next idea.”

Use short sentences and active voice. Avoid unsupported claims.

Examples demonstrate tone, structure and level of detail.

Principle 6: refine instead of restarting



“Keep the layout, but increase contrast and make the headline easier to read.”

Visual prompt: create a design

IMAGE PROMPT

Create a social poster for a beginner AI workshop.
Audience: students and first-time AI users.
Canvas: portrait 4:5.
Style: modern tech, navy background, cyan accents, clean grid.
Headline: “LEARN AI PROMPTING”.
Include 3 short benefits and a “Register” button.
Leave blank space for date, venue and logo.
Keep text readable; avoid clutter.

SUBJECT ✦ PURPOSE

What should be made and why?

AUDIENCE ✦ STYLE

Who is it for, and how should it look?

CONSTRAINTS

What must be included or avoided?

What deficiency looks like in design prompting

PROMPT

"Make a cool flyer."

POSSIBLE RESULT

Generic visuals, wrong colors, weak hierarchy, missing details or too much random text.

FIX

Specify audience, event, exact copy, canvas, style, hierarchy and exclusions.

Deficiency = information the AI has to guess.

Visual prompt: create a short video

VIDEO PROMPT

Create a 6-second cinematic clip for a student tech brand.
Scene: laptop on a desk in a modern learning space.
Action: camera slowly pushes toward the screen; cyan reflections move subtly.
Camera: smooth, steady, medium-wide framing.
Mood: curious, optimistic, focused.
Format: 16:9, no dialogue, no visible logos.
Keep the subject consistent.

SUBJECT ✦ ACTION

What is seen and what changes?

CAMERA ✦ MOOD

How should the shot feel and move?

DURATION ✦ FORMAT

How long and where will it play?

Visual prompt: write content

WRITING PROMPT

Act as a friendly learning facilitator.
Write a 120-word WhatsApp announcement for a beginner AI class.
Mention 3 things learners will learn and end with a clear call to action.
Tone: warm, energetic and simple.
Do not invent date, time or venue; use placeholders.

ROLE ✦ AUDIENCE

Sets voice and reading level.

LENGTH ✦ CONTENT

Sets scope and must-have details.

FACTUAL BOUNDARY

Prevents made-up logistics.

Visual prompt: ask AI for code

BEGINNER CODING PROMPT

Act as a patient web development tutor.
Create a simple HTML page with a heading, paragraph and button.
Explain what each tag does in beginner-friendly language.
Return code first, then explain each line.
Use no external libraries.
Explain how to run and test it.

SPECIFY THE GOAL

Describe the feature or bug.

SPECIFY THE LEVEL

Ask for beginner explanations.

ASK FOR TESTING

Check output and understand the code.

Visual prompt: learn a difficult topic

LEARNING PROMPT

Teach me APIs as if I am a complete beginner.
Start with a real-life analogy.
Explain key terms in plain English.
Give one example, then ask me 3 questions.
Do not reveal answers until I try.
If I get something wrong, explain the misconception kindly.

SCAFFOLD

Start simple, add complexity gradually.

PRACTICE

Ask questions and give feedback.

ADAPT

Tell AI what you already know.

Applications of AI prompting

WRITING
Emails, captions, summaries

DESIGN
Posters, layouts, mockups

VIDEO / AUDIO
Storyboards, scripts, voiceovers

CODING
Explain code, debug, tests

RESEARCH
Outlines, compare sources

BUSINESS
FAQs, workflows, campaigns

EDUCATION
Tutors, quizzes, flashcards

DATA
Explain charts, formulas

Websites for text, research and ideas

ChatGPT	Writing, brainstorming, explanations and images	chatgpt.com
Claude	Writing, analysis and coding help	claude.ai
Gemini	Writing and Google ecosystem tasks	gemini.google.com
Perplexity	Research-oriented answers with source links	perplexity.ai
NotebookLM	Work with supplied sources and study notes	notebooklm.google.com

Features, free limits and availability may change; verify current access.

Websites for AI-assisted design

Canva AI	Social posts, presentations, posters and video	canva.com
Adobe Firefly	Generate and edit images and design assets	firefly.adobe.com
ChatGPT Images	Generate and edit images using natural language	chatgpt.com
Microsoft Designer	AI-assisted social and visual designs	designer.microsoft.com
Ideogram	Image generation for designed text and graphics	ideogram.ai

Websites for video, slides and creative work

Runway	AI video generation and editing	runwayml.com
Pika	AI video and visual effects	pika.art
Luma Dream Machine	Generate video clips from prompts and references	lumalabs.ai
Gamma	Generate presentations, documents and webpages	gamma.app
CapCut	Video editing with AI-assisted tools	capcut.com

Choose the right tool for the job

Need a written draft?	ChatGPT / Claude / Gemini
Need a designed poster?	Canva AI / Firefly / ChatGPT Images
Need a short AI video?	Runway / Pika / Luma
Need slides quickly?	Gamma / Canva
Need to study from notes?	NotebookLM / ChatGPT
Need code help?	ChatGPT / Claude / Gemini

When AI gives a poor result

TOO GENERIC

Add audience, purpose and specific details.

WRONG FORMAT

State the format and show an example.

TOO LONG

Set a word limit and priorities.

INACCURATE

Verify important claims independently.

VISUAL MESS

Reduce clutter and define hierarchy.

MISSED REQUIREMENT

Repeat must-haves and use a checklist.

Responsible prompting: privacy and truth

- Do not paste passwords, private keys or confidential data unnecessarily.
- Verify facts, dates, statistics, citations and code before relying on them.
- Ask for sources and open them; a citation is not proof by itself.
- Use images, voices and likenesses with permission; respect copyright and platform rules.
- Do not use AI to impersonate, deceive, harass or expose private information.
- AI can sound confident and still be wrong. Human review matters.

Follow your organization’s policies and each tool’s privacy settings.

Class exercise: upgrade this prompt

WEAK PROMPT

Make a video for my brand.

Improve it by adding:

- Brand, audience and goal
- Scene and action
- Duration and aspect ratio
- Mood, camera movement and exclusions

CHALLENGE **Write a 3–5 sentence prompt. Swap with a partner: can they picture the result?**

Exercise answer: one stronger version

EXAMPLE PROMPT

Create a 7-second, 16:9 introduction video for a student technology community. Show a laptop on a clean desk in a modern learning space, with subtle cyan accents. Use a slow camera push-in and an optimistic, focused mood. Leave negative space for a title to be added during editing. No visible logos, extra text or rapid camera movement.

Why stronger? Audience ✦ subject ✦ action ✦ duration ✦ format ✦ mood ✦ constraints.

Before you submit a prompt: quick checklist

- □ Did I clearly state the task?
- □ Did I include relevant context and audience?
- □ Did I list must-have details?
- □ Did I specify tone, style or output format?
- □ Did I add constraints or things to avoid?
- □ Did I say how success should be judged?
- □ Will I review and refine the result?

Remember the 5-part beginner formula

GOAL

What do I want?

CONTEXT

Who / why?

DETAILS

What must be included?

FORMAT

What should it look like?

REFINE

What needs improvement?

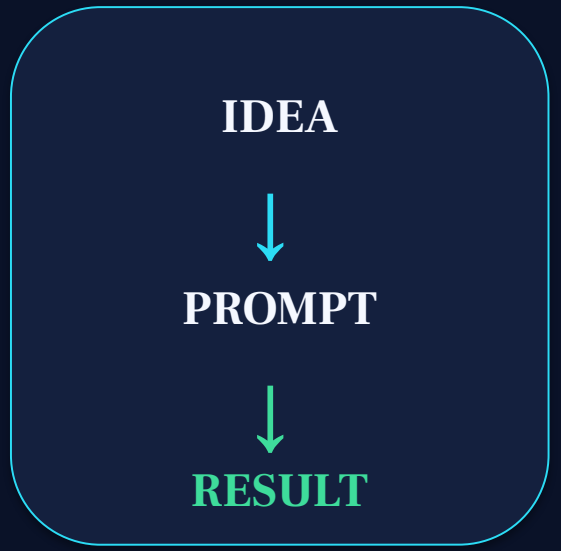
Clear goal ✦ useful context ✦ specific details ✦ chosen format ✦ review

The best prompt communicates the real task clearly.

Official learning resources

OpenAI: How to create a good prompt	help.openai.com/en/articles/4 936 848-how-do-i-create-a-good-prompt-for-an-ai-model
OpenAI Academy: Prompting	academy.openai.com
OpenAI: Image prompting guide	developers.openai.com/api/docs/guides/image-prompting
Runway: Introduction to prompting	help.runwayml.com
Canva AI: Magic Design help	canva.com/en_gb/help/using-magic-design

YOUR IDEA.
YOUR PROMPT.
YOUR NEXT CREATION.



Start simple. Be specific. Check the output. Improve one thing at a time.

Thank you • Web3Nova